NE511 Computational Project Notes

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Contents

1 Multiphysics Problem Description

1.1 Multigroup neutron transport equation with Backward Euler time discretization

The problem described here is for 1-dimensional multigroup neutron transport, coupled to 1-dimensional advection-diffusion through temperature- and flux-dependent cross sections.

The one-dimensional time-dependent neutron transport equation (NTE) for prompt neutrons with isotropic scattering and isotropic source is

$$\left[\frac{1}{v(E)}\frac{\partial}{\partial t}+\mu\frac{\partial}{\partial x}+\sigma_t(x,E)\right]\psi(x,\mu,E,t)=$$
$$\frac{\chi}{2}\int_0^\infty dE'\,(1-\beta(E'))\nu(x,E')\sigma_f(x,E')\phi(x,E',t)+\frac{1}{2}\int_0^\infty dE'\,\sigma_s(x,E'\to E)\phi(x,E',t)+q(x,E,t),$$
$$x\in[0,X],\ E\in[0,\infty),\ \mu\in[-1,1],\ t>0 \quad (1a)$$

$$\phi(x,E,t)=\int_{-1}^{1}d\mu\,\psi(x,\mu,E,t) \quad (1b)$$

$$\psi(x,\mu,E,0)=\psi^0(x,\mu,E) \quad (1c)$$

$$\psi(0,\mu,E,t)=\psi^+(\mu,E,t),\quad \mu>0 \quad (1d)$$

$$\psi(X,\mu,E,t)=\psi^-(\mu,E,t),\quad \mu<0. \quad (1e)$$

Using the standard multigroup approximation, the equation for group g becomes

$$\left[\frac{1}{v_g}\frac{\partial}{\partial t}+\mu\frac{\partial}{\partial x}+\sigma_{t,g}\right]\psi_g(x,\mu,E,t)=$$
$$\frac{\chi_g}{2}\sum_{g'=1}^{N_g}(1-\beta_{g'})\nu\sigma_{f,g'}(x)\phi_{g'}(x,t)+\frac{1}{2}\sum_{g'=1}^{N_g}\sigma_{s,g'\to g}(x)\phi_{g'}(x,t)+q_g(x,t),$$
$$g=1,\ldots,N_g,\ x\in[0,X],\ \mu\in[-1,1],\ t>0 \quad (2a)$$

$$\phi_g(x,t)=\int_{-1}^{1}d\mu\,\psi_g(x,\mu,t) \quad (2b)$$

$$\psi_g(x,\mu,0)=\psi_g^0(x,\mu) \quad (2c)$$

$$\psi_g(0,\mu,t)=\psi_g^+(\mu,t),\quad \mu>0 \quad (2d)$$

$$\psi_g(X,\mu,t)=\psi_g^-(\mu,t),\quad \mu<0. \quad (2e)$$

The Backward Euler temporal discretization scheme is

$$\left.\frac{\partial}{\partial t}\psi(t)\right|_{t=t_{j+1}}\approx\frac{\psi(t_{j+1})-\psi(t_j)}{\Delta t_j} \quad (3)$$

where $t_{j=0}=0$ and

$$\Delta t_j=t_{j+1}-t_j. \quad (4)$$

Substituting this into the multigroup neutron transport equation yields the effective steady-state problem at each time step,

$$\left[\mu\frac{\partial}{\partial x}+\hat{\sigma}_{t,g}^{(j+1)}\right]\psi_g^{(j+1)}(x,\mu,E)=$$
$$\frac{\chi_g}{2}\sum_{g'=1}^{N_g}(1-\beta_{g'})\nu\sigma_{f,g'}^{(j+1)}(x)\phi_{g'}^{(j+1)}(x)+\frac{1}{2}\sum_{g'=1}^{N_g}\sigma_{s,g'\to g}^{(j+1)}(x)\phi_{g'}^{(j+1)}(x)+\hat{q}_g^{(j+1)}(x,\mu),$$
$$g=1,\ldots,N_g,\ j=1,\ldots N_j,\ x\in[0,X],\ \mu\in[-1,1] \quad (5a)$$

$$\phi_g^{(j+1)}(x)=\int_{-1}^{1}d\mu\ \psi_g^{(j+1)}(x,\mu) \quad (5b)$$

$$\psi_g^{(0)}(x,\mu)=\psi_g^0(x,\mu) \quad (5c)$$

$$\psi_g^{(j+1)}(0,\mu)=\psi_g^+(\mu,t_{j+1}),\quad \mu>0 \quad (5d)$$

$$\psi_g^{(j+1)}(X,\mu)=\psi_g^-(\mu,t_{j+1}),\quad \mu<0 \quad (5e)$$

where the augmented total cross section and source are

$$\hat{\sigma}_{t,g}^{(j+1)}=\left(\sigma_{t,g}^{(j+1)}(x)+\frac{1}{v_g\Delta t_j}\right) \quad (6)$$

$$\hat{q}_g^{(j+1)}(x,\mu)=\left(q_g(x,t_{j+1})+\frac{1}{v_g\Delta t_j}\psi_g^{(j)}(x,\mu)\right). \quad (7)$$

1.2 Precursor balance equations

In one dimension, the precursor balance equation is

$$\frac{\partial}{\partial t}C_{d,k}(x,t)=-\lambda_k C_{d,k}(x,t)+\sum_{g=1}^{N_g}\beta_{k,g}(x,t)\nu\sigma_{f,g}(x)\phi_g(x,t),\qquad k=1,\ldots,6,\ t>0 \quad (8a)$$

$$C_{d,k}(x,0)=C_{d,k}^0(x). \quad (8b)$$

With the Backward Euler temporal discretization, this may be written, after some algebra, as

$$C_{d,k}^{(j+1)}(x)=\frac{\frac{C_{d,k}^{(j)}}{\Delta t_j}+\sum_{g=1}^{N_g}\beta_{g,k}\nu\sigma_{f,g}\phi_g^{(j+1)}}{\lambda_k+\frac{1}{\Delta t_j}}. \quad (9)$$

In the neutron transport equation, the delayed neutron source is

$$q_{del,g}(x,\mu)=\frac{1}{2}\sum_{k=1}^{6}\lambda_k\chi_{g,k}^d C_{d,k}, \quad (10)$$

so, substituting eq. (9) into eq. (10), the time-discretized delayed neutron source in the transport equation becomes

$$q_{del,g}(x,\mu) = \frac{1}{2}\sum_{g'=1}^{N_g} \Gamma_{g'\to g}^{(j+1)} \nu\sigma_{f,g'}\phi_{g'}^{(j+1)} + \overline{q}_{del,g}^{(j+1)}(x,\mu) \tag{11}$$

where

$$\Gamma_{g'\to g}^{(j+1)} = \sum_{k=1}^{6} \chi_{g,k} \frac{\lambda_k \Delta t_j \beta_{g',k}}{\lambda_k \Delta t_j + 1} \tag{12}$$

$$\overline{q}_{del,g}^{(j+1)}(x,\mu) = \frac{1}{2}\sum_{k=1}^{6} \lambda_k \chi_{g,k} \left(\frac{C^{(j)}}{\lambda_k \Delta t_j + 1} \right). \tag{13}$$

Here, $\overline{q}^{(j+1)}$ represents the decay of precursors that existed in time step j, and $\Gamma_{g'\to g}^{(j+1)}$ represents the proportion of fissions in group g', during the time step Δt_j, whose delayed neutrons decay during the same time step (immediate precursor decay).

1.3 Equivalent steady-state transport problem with delayed neutrons

Adding eq. (11) to eq. (5), the quasi-steady state transport equation to be solved at each time step, including the effect of delayed neutrons, is

$$\begin{aligned} \left[\mu\frac{\partial}{\partial x} + \hat{\sigma}_{t,g}^{(j+1)}\right] \psi_g^{(j+1)}(x,\mu,E) = \\ \frac{\chi_g}{2}\sum_{g'=1}^{N_g}(1-\beta_{g'})\nu\sigma_{f,g'}^{(j+1)}(x)\phi_{g'}^{(j+1)}(x) + \frac{1}{2}\sum_{g'=1}^{N_g}\left(\sigma_{s,g'\to g}^{(j+1)}(x) + \Gamma_{g'\to g}^{(j+1)}\nu\sigma_{f,g'}\right)\phi_{g'}^{(j+1)}(x) \\ + \hat{q}_g^{(j+1)}(x,\mu) + \overline{q}_{del,g}^{(j+1)}(x,\mu), \\ g = 1,\ldots,N_g,\ j = 1,\ldots N_j,\ x\in[0,X],\ \mu\in[-1,1] \end{aligned} \tag{14a}$$

$$\phi_g^{(j+1)}(x) = \int_{-1}^{1} d\mu\ \psi_g^{(j+1)}(x,\mu) \tag{14b}$$

$$\psi_g^{(j+1)}(0,\mu) = \psi_g^{+}(\mu, t_{j+1}), \quad \mu > 0 \tag{14c}$$

$$\psi_g^{(j+1)}(X,\mu) = \psi_g^{-}(\mu, t_{j+1}), \quad \mu < 0 \tag{14d}$$

where

$$\psi_g^{(0)}(x,\mu) = \psi_g^{0}(x,\mu), \qquad g = 1,\ldots,N_g \tag{15a}$$

$$C_{d,k}^{(0)}(x) = C_{d,k}^{0}(x), \qquad k = 1,\ldots,6 \tag{15b}$$

After solving for the neutron distribution, the precursor distribution can be recovered using eq. (9).

1.4 Multigroup Low-Order Quasidiffusion Equations

Taking the zeroth and first spatial moments ($\int \cdot \, d\mu$ and $\int \mu \cdot d\mu$) of eq. (14) (omitting $j+1$ index) yields

$$\frac{\partial}{\partial x} J_g(x) + \sigma_{t,g}\phi_g(x) = \chi_g \overline{(1-\beta)\nu\sigma_f}\phi + \sum_{g'=1}^{N_g}\left(\sigma_{s,g'\to g} + \Gamma_{g'\to g}\nu\sigma_{f,g'}\right)\phi_{g'}(x) + \hat{Q}_g(x) + \overline{Q}_{del,g} \quad (16a)$$

$$\frac{\partial}{\partial x}(E_g\phi_g) + \sigma_{t,g}J_g = \tilde{q}_g \quad (16b)$$

$$J_g(0) = C^b_{0,g}\phi_g(0), \qquad J_g(X) = C^b_{X,g}\phi_g(X) \quad (16c)$$

where the initial conditions are derived from taking the zeroth and first spatial moments of ψ^0 and

$$\hat{Q}_g(x) = \left(2q_g(x) + \frac{1}{v_g\Delta t_j}\phi_g^{(j)}(x)\right) \quad (17)$$

$$\overline{Q}_{del,g} = \sum_{k=1}^{6}\lambda_k\chi_{g,k}\left(\frac{C^{(j)}}{\lambda_k\Delta t_j + 1}\right) \quad (18)$$

$$\tilde{q}_g = \frac{1}{v_g\Delta t_j}J^{(j)}(x) \quad (19)$$

and

$$\overline{(1-\beta)\nu\sigma_f} = \frac{\sum_{g=1}^{N_g}(1-\beta_g)\nu\sigma_{f,g}}{\sum_{g=1}^{N_g}\phi_g} \quad (20)$$

$$\phi = \sum_{g=1}^{N_g}\phi_g. \quad (21)$$

The nonlinear closure term (Eddington Factor) is

$$E_g = \frac{\int\limits_{-1}^{1}\mu^2\psi_g d\mu}{\int\limits_{-1}^{1}\psi_g d\mu} \quad (22)$$

1.5 Advection-diffusion equation

The advection-diffusion equation in 1D is

$$\left[\frac{\partial}{\partial t} - \frac{\partial}{\partial x}k_b\frac{\partial}{\partial x} + \dot{m}c_p\frac{\partial}{\partial x}\right]T(x,t) = W_f f(x,t) \quad (23)$$

or in operator form,

$$\mathcal{H}T(x,t) = Q_f \tag{24}$$

with appropriate boundary conditions. the fission reaction rate f is

$$f(x) = \sum_{g=1}^{N_g} \sigma_{f,g}\phi_g(x,t) \tag{25}$$

and

$$\begin{aligned} c_p &= \text{specific heat capacity} \\ k_b &= \text{turbulent diffusion coefficient} \\ \dot{m} &= \text{mass flow rate} \\ W_f &= \text{fission energy deposition} \end{aligned}$$

Discretizing with Backward Euler, this equation becomes

$$\left[\frac{1}{\Delta t_j} - \frac{\partial}{\partial x}k_b\frac{\partial}{\partial x} + \dot{m}c_p\frac{\partial}{\partial x}\right]T^{(j+1)}(x) = W_f f^{(j+1)}(x) + \frac{1}{\Delta t_j}T^{(j)} \tag{26}$$

$$\hat{\mathcal{H}}T^{(j+1)}(x) = \hat{Q}_f^{(j+1)} \tag{27}$$

which is an equivalent steady-state advection-diffusion problem at each time step with a local removal term.

1.6 Cross-section feedback

For this study, the macroscopic feedback will be modeled with

$$\sigma_u(x) = \sigma_{u,0} + \sigma_{u,1,d}\left(\sqrt{T(x) + \kappa W_f f(x)} - \sqrt{T_{d,0}}\right) + \sigma_{u,1,b}\left(T(x) - T_0\right) \tag{28}$$

for each reaction u, where $\kappa, \sigma_{u,0}, \sigma_{u,1,d}, \sigma_{u,1,b}, T_{d,0}$, and T_0 are material constants, defined for each neutron energy group g.

2 MNP Method

2.1 Basic method details

2.2 Iterative Algorithms

3 NILO-MNP method

3.1 Method details

3.2 Iterative Algorithm

Appendix A LD Discretization of Neutron Transport equation

Appendix B FV Discretization of advection-diffusion equation

Appendix C LD Discretization of QD equations